

1.1 Classwork: Measurement, Precision, and Significant Figures

Reference (provided). 1 in. = 2.54 cm 1 m $\approx$ 3.28 ft 1 h = 3600 s
1 km = 1000 m 1 m = 100 cm 1 cm = 10 mm 1 kg = 1000 g

1. Write down the number of significant digits in each measurement.

- (a) 42 2 (c) 0.0058 2 (e) 250.0 4
(b) 3.60 3 (d) 0.01070 4 (f) 6.022 4

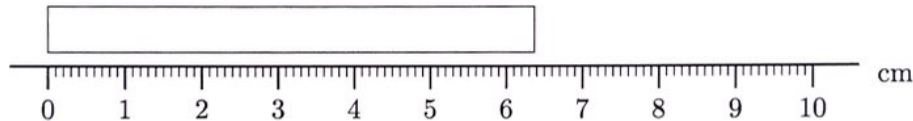
2. Round each value to three significant figures. Copy the calculator display followed by three dots, then round:

$$\sqrt{2} = 1.4142135 \dots \approx 1.41$$

- (a) The population of Milan: 1,371,498 (b) $\sqrt{5}$

$$1,371,498 \approx 1,370,000 \quad \sqrt{5} = 2.236067 \dots \approx 2.24$$

3. The bar below is measured with a centimeter ruler marked in millimeters.



- (a) Record the length of the bar. Include one estimated digit — the last digit, the one you are not certain of.

6.38 cm

- (b) How many significant figures does your measurement have?

3

- (c) Write the same length in millimeters, and again in meters.

63.8 mm 0.0638 m

4. Write in scientific notation, rounded to three significant figures.

- (a) The distance from New York to Rome: 6,890,000 meters. (b) The diameter of a red blood cell: 0.0000078 meters.

$$6.89 \times 10^6 \text{ m}$$

$$7.80 \times 10^{-6} \text{ m}$$

5. Write each value as an ordinary decimal number.

(a) 4.05×10^5 grams

$$405,000 \text{ g}$$

(b) 2.9×10^{-4} seconds

$$0.00029 \text{ s}$$

6. Convert between units, rounding to three significant figures. Write the conversion as a fraction and show the units cancelling.

(a) The Colosseum stands 48.5 meters tall. What is its height in feet?

$$48.5 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} = 159.12 \text{ ft} \\ (159. \text{ ft})$$

(b) A 12 inch ruler is how many centimeters long?

$$12 \text{ in} \times \frac{2.54 \text{ cm}}{1 \text{ in}} = 30.48 \text{ cm (exact)}$$

7. A bus on Via del Corso travels at 30 kilometers per hour. What is this speed in meters per second?

$$30 \frac{\text{km}}{\text{hr}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{3600 \text{ s}} \approx 8.33 \text{ m/s}$$

8. The mass of the Earth is 5.972×10^{24} kg and the mass of Jupiter is 1.898×10^{27} kg. How many times more massive is Jupiter than the Earth? Round to three significant figures.

$$\frac{1.898 \times 10^{27}}{5.972 \times 10^{24}} = 0.317816... \times 10^3 \\ \approx 318 \text{ times}$$

9. Challenge: Estimate the number of seconds you have been alive. Give your answer in scientific notation, and explain why it would be wrong to report this number to nine significant figures.

$$68 \text{ y} \cdot \frac{365 \text{ d}}{1 \text{ y}} \times \frac{24 \text{ h}}{1 \text{ d}} \times \frac{3600 \text{ s}}{1 \text{ h}} = 2.14 \times 10^9 \text{ s}$$